

13.1.1. Fibonacci Sequence using Modules

Algorithm: Fibonacci Sequence using Modular Programming

Part 1: Module (fibonacci_module.py)

1. **Start module**
 2. **Define function**
 - Function name: generate_fibonacci_sequence(max_value)
 3. **Initialize variables**
 - Set $a = 0$, $b = 1$
 - Create empty list result = []
 4. **Generate Fibonacci sequence**
 - While $a \leq \text{max_value}$:
 - Add a to result
 - Update values:
 - $a = b$
 - $b = a + b$ (use previous values logic)
 5. **Return result list**
 6. **End module**
-

Part 2: Main Program (main.py)

1. **Start**
2. **Import module**
 - Import fibonacci_module
3. **Input**
 - Read integer n
4. **Validate input**
 - If $n \leq 0$:

- Print "Please enter a positive integer"

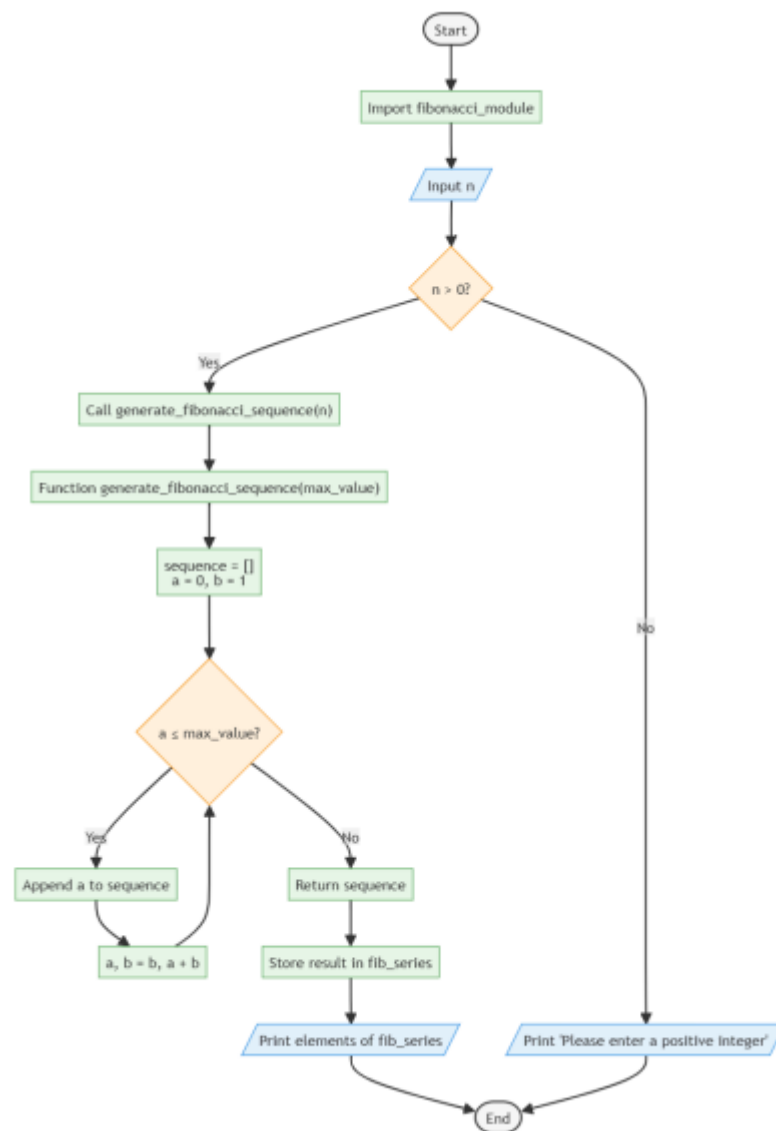
5. Else (valid input)

- Call function:
 - `fib_series = generate_fibonacci_sequence(n)`

6. Display output

- Convert list to string
- Print numbers space-separated

7. End



Flowchart :

Programme :

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12.1.1. Fibonacci Sequence using Modules

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Module Requirements

- Create a module `fibonacci_module.py` with a function `generate_fibonacci_sequence(max_value)` that returns list of all Fibonacci numbers less than or equal to the maximum value.

Main Program Requirements

Create `main.py` that performs the following:

- Import the `fibonacci_module`.
- Read an integer `n` from user input.
- Validate if `n` is positive.
- If valid, call the module function and display the Fibonacci sequence.
- If invalid, display an error message.

Input Format:

- Single integer `n` representing the maximum value for Fibonacci sequence.

Output Format:

- If `n > 0`, print the Fibonacci sequence as space-separated numbers.
- If `n ≤ 0`, print "Please enter a positive integer".

Note:

- Two files are provided: `fibonacci_module.py` and `main.py`. The `main.py` file handles input and output. You need to complete both files based on the question requirements.

Sample Test Cases

main.py

```
1 n = int(input())
2
3
4 if n > 0:
5     fib_series = fibonacci_module.generate_fibonacci_sequence(n)
6     print(' '.join(map(str, fib_series)))
7 else:
8     print("Please enter a positive integer")
9
10 def generate_fibonacci_sequence(max_value):
11     a, b = 0, 1
12     result = []
13     while a <= max_value:
14         result.append(a)
15         a, b = b, a + b
16     return result
17
18 n = int(input())
19 if n <= 0:
```

Average time

Maximum time

0.006 s

0.008 s

6.13 ms

8.00 ms

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1

Expected output

0 1 1 2 3 5 8

Actual output

0 1 1 2 3 5 8

Test case 2

Test case 3

Terminal

Test cases

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